

FVN Research

Reproducibility Runbook

Command sequence and replay logic for the FVN Archive 001 research system.

Core reproduction commands

Command	Purpose
make run-e2e-smoke	Run deterministic end-to-end smoke pipeline
make check-live-readiness	Validate live data/compliance prerequisites
make run-live-pipeline-dry	Plan live commands behind readiness gate
make build-data-lineage-graph	Generate nodes, edges, and lineage map
make validate-schema-contracts	Validate critical artifact schemas
make build-reproducibility-pack	Build audit bundle and checksums
make build-release-checklist	Build pre-publication release gate
make build-publication-package	Build sanitized public package
make build-final-archive-freeze	Build immutable library freeze metadata

Minimum replay path

```
make run-e2e-smoke
make build-data-lineage-graph
make validate-schema-contracts
make build-reproducibility-pack
make build-release-checklist
make build-publication-package
make build-final-archive-freeze
```

Live-data path

Live execution is intentionally gated. The archive should not download or process live source data unless compliance and source prerequisites pass.

```
make check-live-readiness
make run-live-pipeline-dry
make run-live-pipeline
```

Replay interpretation

- Smoke replay proves that the pipeline layers can execute end-to-end on deterministic synthetic data.

- Schema contracts prove that critical artifacts match expected structure and quality gates.
- Lineage outputs show how raw inputs, commands, processed artifacts, reports, and audit outputs connect.
- Freeze metadata fixes the canonical version, checksums, and library storage record.